On maximum likelihood estimation of intensity function for Poisson process with deep learning

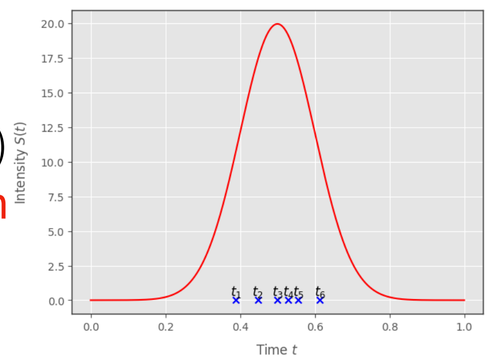
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Introduction

Poisson Process

- A mathematical model describing the random occurrence of events in space and time.
- Definition of a Poisson process X on $[0, 1]$.
  - X is a non-negative integer-valued random measure on $[0, 1]$ (counting the number of events)
  - $X(\{0\}) = 0,\ a.s.$
  - For any $0 \leq t_1 < \cdots < t_N \leq 1$, $X([t_1, t_2)), \ldots, X([t_{N-1}, t_N))$ are independent.
  - For any $0 \leq s < t \leq 1$, $X([s, t)) \sim \mathrm{Poisson}(\int_s^t S(\tau)\, d\tau)$

Intensity function



Characterizes the occurrence rate of events at each time point

- In practice, the intensity function is not fixed; it varies with underlying factors.
- To capture this, we model the intensity function as depending on covariates Z:

"Conditional" intensity function

$$X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$$

Challenges in the Theoretical Analysis of the Nonparametric MLE

- $A \subset \mathbb{R}^{d_1}$: space of events, $C \subset \mathbb{R}^{d_2}$: space of covariates
- i.i.d. observations of events and covariates:

$$Z_i \sim_{\text{i.i.d.}} P_Z, \quad X_i \sim \text{Poisson process}(S_0(\cdot \mid Z_i)), \quad i = 1, \ldots, n$$

- Negative log likelihood

$\boldsymbol{x}_{ij}$: events corresponding to X_i
$N_i = X_i(A)$, ν: measure on A

$$\ell_n(S) := \frac{1}{n} \sum_{i=1}^{n} \left\{ -\sum_{j=1}^{N_i} \log(S(\boldsymbol{x}_{ij} \mid \boldsymbol{z}_i)) + \int_A S(\boldsymbol{x} \mid \boldsymbol{z}_i)\, d\nu(\boldsymbol{x}) \right\}$$

- $\hat{S}_n \in \arg\min_{S \in \mathcal{F}_n} \ell_n(S)$: MLE, $\mathcal{F}_n$: Hypothesis space
- The excess risk of the MLE involves the KL divergence.
- **But the KL divergence diverges easily ⇒ Conventional generalization error analysis is challenging 😞**

Our Research Goal

- **Understand the theoretical properties of the MLE.**
  - **Evaluating the Hellinger distance instead of KL, we obtain an oracle inequality under mild conditions.**

Theory & Experiments

Notation and Assumptions

- $\tilde{H}^2(S_1, S_2) := \mathbb{E}_Z \left[\frac{1}{2} \int_A (\sqrt{S_1(\boldsymbol{x} \mid Z)} - \sqrt{S_2(\boldsymbol{x} \mid Z)})^2 d\nu(\boldsymbol{x}) \right]$
  - **Evaluate this quantity instead of the KL divergence.**
- We consider a DNN of depth L, width r per layer, whose parameters are bounded in absolute value by B.
  - **Every layer, including the output layer, uses the ReLU activation function.**
- Assumption：

$$\exists c_0 > 0,\ \forall n \in \mathbb{N},\ \exists S_n^* \in \mathcal{F}_n,\ \forall \boldsymbol{x} \in A,\ \forall \boldsymbol{z} \in C;\ \frac{S_0(\boldsymbol{x} \mid \boldsymbol{z})}{S_n^*(\boldsymbol{x} \mid \boldsymbol{z})} \leq c_0^2$$

  - With DNN-based estimation, this condition is automatically met.

Oracle Inequality

Theorem (Oracle Inequality)

We consider the estimation of the conditional intensity function of a Poisson process. Suppose that δ_n and a universal constant c satisfies the following inequality:

$$\sqrt{n}\delta_n^2 \geq c\Psi(\delta_n)$$

Then, for any $\delta \geq \delta_n$, the following inequality holds.

Variance

$$\mathbb{E}[\tilde{H}^2(\hat{S}_n, S_0)] \leq 514(c_0 + 1)^2 \left(\underline{\delta^2} + \frac{c^3}{n} + \underline{\frac{\tilde{H}^2(S_n^*, S_0)}{\text{Bias}^2}} \right)$$

- $\Psi(\delta)$ **corresponds to the complexity of the model.**
- If we can evaluate both the complexity and the bias, then the convergence rate of the MLE can be determined!

Convergence Rates in Deep Learning

- By applying the oracle inequality, we can derive convergence rates in deep learning.

The function space	$\mathcal{C}^\beta(\mathcal{M} \times C, Q)$	$\mathcal{G}_{\text{comp}}$
Convergence rates	$-\frac{\beta}{\beta + d_{\mathcal{M}} + d_2}$	$-\max_{i=0,\ldots,q} \frac{\beta_i^*}{\beta_i^* + t_i}$

- **Deep learning can appropriately capture the underlying low-dimensional structure.**

Numerical Experiments

